

Sonica Jose Gomez

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PROFILE

- Detail-oriented B.Tech graduate with a foundation in Python, SQL, Java, and Machine Learning, experienced in developing end-to-end applications across NLP, supervised learning, computer vision, and signal processing.
- Strong analytical and problem-solving skills with hands-on experience in application development, model evaluation, and deployment.
- Combines an Electronics & Communication background with software and ML skills to develop practical, data-driven solutions to real-world problems.

TECHNICAL SKILLS

- **Programming:** Python, SQL, Java, C, HTML.
- **Machine Learning:** Naive Bayes, KNN, Logistic Regression, SBERT, Deep Learning, MediaPipe, EDA.
- **Software Engineering:** Streamlit, PostgreSQL, Supabase, Git, Relational Database Design.
- **Vision & Hardware:** OpenCV, Arduino Nano, BioAmp, EOG Signal Acquisition, Digital Signal Processing

PROFESSIONAL SKILLS

- Structured Thinking | Technical Documentation | Attention to Detail | Complex Problem Solving.

EDUCATION

- B.Tech | Electronics and Communication, Sree Chithra Thirunal College of Engineering | Graduated: 2026
- Minor in Machine Learning | CGPA: 8.4
- 12th | Sarvodaya Vidyalaya | Percentage: 99%
- 10th | Sarvodaya Vidyalaya | Percentage: 90%

PROJECTS

CreditWise | Data Driven Approval Prediction System | <https://credit-wise2.streamlit.app/>

- Developed an end-to-end supervised ML system to predict whether a loan application would be approved or rejected for initial applicant screening.
- Performed EDA, preprocessing, feature engineering, and model comparison using KNN, Logistic Regression and Naive Bayes.
- Evaluated models using Precision, Recall, and F1-score, prioritizing Precision to reduce the risk of incorrectly approving high-risk applicants.
- Selected Naive Bayes based on stronger Precision performance and deployed the trained prediction pipeline through Streamlit.

Skillsync | Skill Gap Analyzer Platform | : <https://skillsync-ai.streamlit.app/>

- Addressed the challenge of aligning resumes with complex job descriptions by developing an AI-powered skill-gap and resume matching platform.
- Implemented SBERT-based semantic similarity to evaluate resumes against job descriptions, identify relevant/missing skills, and generate ATS-oriented scores.
- Developed the web application using Streamlit and integrated Supabase for structured backend database storage.
- Built and deployed an end-to-end platform that provides skill-gap analysis and targeted learning recommendations based on identified gaps.

Advanced Eye Gaze Typing System | Assistive Technology:

- Designed an assistive communication system to enable individuals with severe motor limitations to communicate through intentional eye movements.
- Built the EOG signal-acquisition setup using Arduino Nano, BioAmp Module and gel electrodes, capturing electrical activity generated by eye movements.
- Utilized the BioAmp's integrated low-pass, high-pass, and band-pass filtering capabilities for EOG signal acquisition and signal conditioning.
- Integrated Python and MediaPipe for eye/blink tracking and developed a functional virtual keyboard that converts intentional eye movements into digital keyboard inputs.

WORK EXPERIENCE

Communication Navigation & Surveillance Intern | Airport Authority Of India | May 2025 - Jun 2025

- Studied the practical application of RF communication within aviation communication and surveillance networks.
- Examined the operational principles and accuracy requirements of active radar systems used for tracking.
- Studied aviation tracking technologies including ADS-B and DME, focusing on their communication and navigation functions.
- Developed foundational understanding of RF signaling, tracking systems, signal precision, and safety-critical aviation operations.

Web Developer Intern | Teknokraaft Info Systems Pvt. Ltd. | July 2024 – Aug 2024

- Learned foundational web development concepts and programming practices during the initial phase of the internship.
- Applied basic HTML markup and semantic structuring to organize digital information effectively.
- Developed static webpages using HTML, with a focus on structured and readable data tables.
- Gained hands-on experience with webpage layout organization and structured data presentation.

CERTIFICATIONS

- Microsoft Certified: Azure Cloud Fundamentals (AZ-900) | In progress | Expected completion August 2026
- Google AI Professional Certificate: Coursera (2026)
- Introduction to Machine Learning : IIT Madras (2025)
- Introduction to SQL : Simplilearn (2025)
- Deep Learning: IIT Kharagpur: (2025)